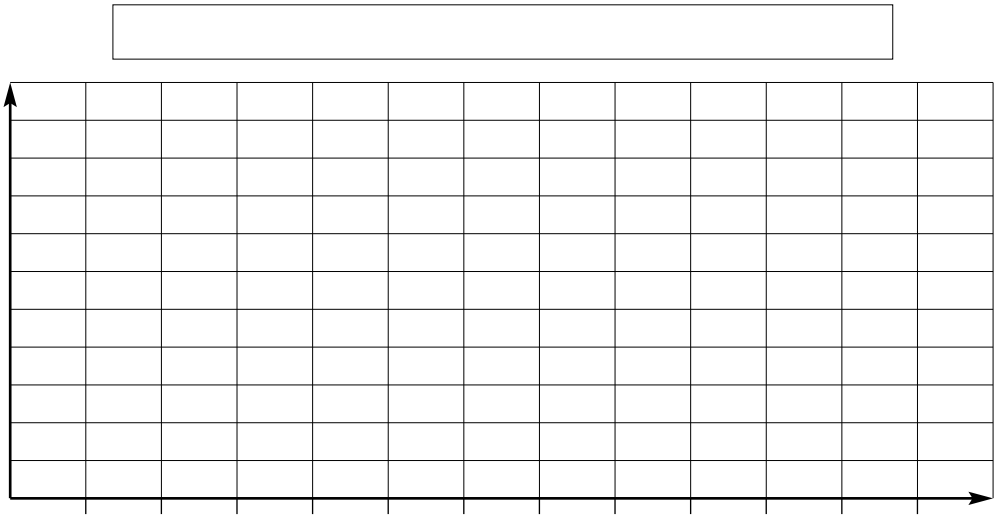
Hearing range and sound range

12a

Animals can hear sounds that are low pitch (i.e. low frequency) and also high pitch (high frequency). But the sounds they make often have a much smaller range.

Animal	Hearing range	Sound production range
Bat	2 to 110 kHz	
Beluga whale	100 Hz to 120 kHz	100 Hz to 120 kHz
Cat	70 Hz to 60 kHz	
Chicken	100 to 2 000 Hz	
Dog	50 to 50 000 Hz	
Elephant	1 to 12 000 Hz	5 to 9 000 Hz
Goldfish	20 to 3 000 Hz	
Human	20 Hz to 20 kHz	0.1 to 5 kHz
Mouse	1 to 100 kHz	
Moth	1 to 240 kHz	
Porpoise	1 to 150 kHz	1 to 150 kHz

1 Make a bar graph that shows the hearing range of the various animals. The horizontal axis represents the frequencies and the vertical axis lists the animals. Write a title



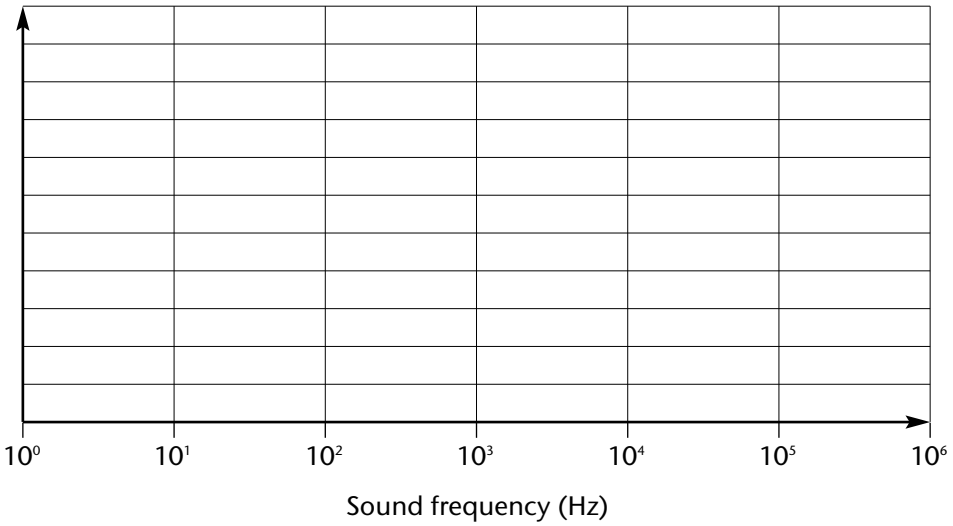
2 The low end of the frequency ranges cannot be read from the above graph. The graph on the next page uses an exponential scale for frequency. The first scale mark represents 10^0 (= 1), the second 10^1 (= 10), the third 10^2 (= 100) etc. Plot the hearing frequency range for the various animals using the exponential scale for frequencies. Add different coloured bars to indicate sound production ranges.

Key: blue = hearing range
red = sound production range

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Hearing range and sound range

12b



3 Consider the animals listed on the previous page. Sort them by size.

Largest _____

_____ smallest

4 **a** *The bigger the skull, the lower the pitch an animal can hear.* Is this generalisation true? _____

b Make a more accurate generalisation relating animal size to hearing.

_____

_____

c Make another generalisation about the hearing range.

_____

_____

5 Infrasound is the name given to those sounds that are below the lowest limit of human hearing while ultrasound covers sounds that are above the human upper limit. Use the information in the above table and list the animals according to the following headings.

Hears infrasound and sound only	Hears sound and ultrasound only	Hears infrasound, sound and ultrasound

6 **a** A baby growing inside a uterus should not be 'looked at' with an X-ray machine. Explain.

_____

b Rather than use X-rays hospitals use ultrasound (frequency range of 1 to 20 MHz) to produce an image of babies. Rewrite 20 MHz in Hz. _____

c Rewrite 20 MHz in kHz. _____

d State which animals are capable of hearing the sound made by an ultrasound machine.

_____

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